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10 **1. Introduction**

11 Two of the defining challenges of this century compete for the same re-  
12 source: land. The energy transition requires terawatts of new solar pho-  
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one  
14 to two hectares per megawatt. Food production must rise while crops are  
15 exposed to more frequent heat waves and droughts. Warming alone is pro-  
16 jected to reduce global yields of maize, wheat, rice and soybean by several  
17 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation  
18 of climate and crop models shows these impacts emerging earlier than pre-  
19 viously expected (Jägermeyr et al., 2021). Croplands are also where solar  
20 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-  
21 sible and close to rural electricity demand, so converting farmland into solar  
22 farms is tempting and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same  
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,  
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,  
26 a hectare can deliver more food plus energy than the two uses side by side,  
27 with land-equivalent ratios (LER) well above one in field trials and mod-  
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;  
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-  
30 climate. It lowers canopy temperature and evaporative demand, conserves  
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-  
32 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;  
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-  
34 didate adaptation measure against heat and drought stress, as reviewed by  
35 Mamun et al. (2022), in addition to a mitigation measure through renew-  
36 able electricity. The same shade, however, also reduces the light available for  
37 photosynthesis. Regulators increasingly require AV systems to retain most  
38 of the reference crop yield. The German pre-standard DIN SPEC 91434, for  
39 example, requires at least 66% of the reference yield (DIN, 2021), and other  
40 schemes and many farmers set stricter floors. How much electricity can be  
41 produced without sacrificing food security depends on the crop, the local  
42 climate, the year and the growth stage.

43 Most AV designs fix the geometry of the array, i.e. the ground coverage  
44 ratio (GCR, module area per ground area) and the tilt or tracking rule, from

generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which would need very many simulated seasons for global, multi-decadal assessments.

Two methodological strands are relevant. Differentiable process models, which couple physical simulators with learning, allow parameters and policies to be fitted by gradient descent through the simulator (Reichstein et al., 2019; Shen et al., 2023), and we use them for both calibration and control. Decision making under deep uncertainty asks for designs that perform acceptably across plausible models and not only under the best estimate (Marchau et al., 2019), and scenario-based designs have finite-sample guarantees under convexity assumptions that we do not meet (Calafiore and Campi, 2006; Campi and Garatti, 2008); our design rules are heuristic counterparts and we compare them with a simple safety margin.

Two questions follow. First, how much of the electricity that an agrivoltaic system can produce under a yield-retention floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen

83 from its historical seasons only, with the same rule for every method, and  
84 performance is measured on held-out sites and years, over three training seeds  
85 and three spatial splits. The third is a differentiable agrivoltaic simulator  
86 that supports both, coupling hourly solar geometry, single-axis trackers with  
87 backtracking, row-to-row shading and PV electricity, the direct and diffuse  
88 light reaching the crop, canopy temperature and reference evapotranspiration  
89 under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended  
90 with a flowering-stage shade penalty. All components are differentiable tensor  
91 operations (Karniadakis et al., 2021). The PV part is benchmarked against  
92 PVGIS. The crop response to shade is learned from the open data of a meta-  
93 analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato  
94 and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al.,  
95 2023), and propagated through the simulator as an ensemble.

96 We apply the protocol to 400 crop–site pairs that represent where maize,  
97 wheat, rice, soybean and potato are grown worldwide, sampled in proportion  
98 to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr  
99 et al., 2021), and driven by NASA POWER daily weather for 2001–2019  
100 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main  
101 findings are as follows.

- 102 1. Light sharing is the main lever on electricity at a given floor, although  
103 economically it pays only at high prices (item 6). At the same mean  
104 yield retention of 90%, arrays with light-sharing control carry 2.1 to  
105 2.9 times the module area of static arrays and produce 1.8–2.4 times  
106 as much electricity (Section 3.1).
- 107 2. The best hand-written rule follows crop phenology and needs no weather  
108 forecast. Rules that react to heat or drought stress are not better and  
109 depend on forecast quality (Section 3.3).
- 110 3. The crop response to shade can be learned from data. A hierarchical  
111 model fitted to 428 observations from 58 field studies predicts a held-  
112 out study of the four staple crop types to about 0.22 in relative yield,  
113 no better than proportional loss (it is better only for crop types the  
114 simulator does not represent), so between-study spread is large. The  
115 simulator is close to these curves for cereals and potato but milder  
116 than the meta-analysis for maize and soybean, whereas one maize trial  
117 is milder still; the gains for those two crops are therefore uncertain and  
118 possibly overestimated (Section 3.7).
- 119 4. A learned controller adds a small gain over the best rule: 0.044 LER

over the written phenology rule and 0.018 over a rule tuned on the same training seasons. Training it on ensembles of shade responses, or giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.8).

5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under the mean response performs alike at equal compliance (differences within the uncertainty), and no rule is safe outside the ensemble, so the ensemble is useful to size a margin and not to replace it (Section 3.8).
6. Under illustrative costs, light sharing breaks even with the open field at 70 to 92 USD MWh<sup>−1</sup> (static arrays at 59), learning is worth about 2kUSD ha<sup>−1</sup> yr<sup>−1</sup> over a tuned rule at 90 USD MWh<sup>−1</sup>, and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh<sup>−1</sup> (Section 3.9).
7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr<sup>−1</sup>, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.11).

## 2. Materials and methods

### 2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below −8°C, in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,

2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels +1.5, +2 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO<sub>2</sub> concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

## 2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

*Hourly irradiance.* Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

*Array geometry and PV electricity.* We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation  $\phi$  is limited to  $\pm 60^\circ$ . The ground coverage ratio  $g$  is the module width divided by the row pitch and is restricted to  $g \leq 0.5$ . For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let  $\psi$  be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011) is  $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$  when  $\cos \psi < g$ , and  $\phi_{\text{bt}} = \psi$  otherwise. The *light-sharing* rotation  $\phi_{\text{sh}}$  is the feasible angle closest to edge-on

190 to the sun, which minimises the shadow on the crop. Given a light-sharing  
 191 level  $u \in [0, 1]$  chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

192 Plane-of-array irradiance combines three terms: beam irradiance on the mod-  
 193 ule, reduced by the fraction of the module shaded by the neighbouring row,  
 194  $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$ ; isotropic sky diffuse; and ground-reflected  
 195 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),  
 196 with an hourly air temperature interpolated sinusoidally between the daily  
 197 extremes. Electricity per unit land is plane-of-array irradiance  $\times \eta \times g$ , with  
 198  $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$  for module efficiency, temperature  
 199 losses and system losses. The installed capacity is  $2.1 \text{ g MWp ha}^{-1}$ . A con-  
 200 ventional ground-mounted PV plant ( $g = 0.4$ , backtracking, no crop) is the  
 201 energy reference  $E_{\text{PV}}$ .

202 *Validation of the PV model.* We compared the simulated specific yield of an  
 203 isolated tracker row ( $g = 0.05$ , negligible row-to-row shading) with PVGIS 5.2  
 204 (European Commission, Joint Research Centre, 2022), for a horizontal north-  
 205 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29  
 206 returned data). Across sites, the simulated and PVGIS specific yields were  
 207 correlated with  $r = 0.971$ . The simulator was on average 4.9% lower (mean  
 208 absolute deviation 5.5%). This is consistent with its  $\pm 60^\circ$  rotation limit  
 209 and with the different irradiance databases (NASA POWER vs. SARA2,  
 210 ERA5 and NSRDB). The benchmark covers an isolated row at low density  
 211 only; electricity at high density, with row-to-row shading and diffuse-light  
 212 effects, was not validated.

213 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction  
 214 of ground reached by beam light is  $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$ . The  
 215 fraction of sky diffuse reaching the ground is a two-dimensional isotropic  
 216 sky-view factor. We pre-computed it for 25 values of  $g$  and 13 rotation  
 217 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly  
 218 (differentiably). The daily ratio  $s$  of the irradiance at the crop to that of the  
 219 open field defines the crop's shading. The canopy maximum temperature is  
 220 lowered by  $\kappa(1 - s)$ .

221  $\kappa$  is uncertain. Field studies under PV modules report small reductions  
 222 of daytime crop or canopy temperature and little change in air temperature  
 223 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds

calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate  $\kappa = 1^\circ\text{C}$  as the main case and report  $\kappa = 0$  (no cooling) and  $\kappa = 2^\circ\text{C}$  for all methods (Section 3.2). Nocturnal warming under the modules is not represented. The reference evapotranspiration under the modules,  $\text{ET}_0$ , is re-computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998) from the reduced radiation and canopy temperature. Soil water conservation under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018) therefore emerges from the model, but rain redistribution by the modules (Marrou et al., 2013a) is ignored.

*Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019) with the species parameters of its Table 1a (Table 1). Daily biomass growth is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

where  $R_{\text{crop}} = sR$  is the radiation received by the crop. The terms are as follows.

- $f_{\text{Solar}}$  is the canopy’s radiation interception. In SIMPLE it depends only on thermal time, with senescence accelerated by heat and drought stress. Shade therefore reduces the daily biomass increment but not leaf-area development.
- $f_{\text{Heat}}$  decreases linearly between the heat threshold  $T_{\text{heat}}$  and the extreme temperature  $T_{\text{ext}}$  of the canopy maximum temperature.
- $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$ , where the agricultural reference index for drought (Woli et al., 2012) is computed from a daily soil-water balance with curve-number runoff, drainage and root water uptake. Irrigated fields have  $f_{\text{Water}} = 1$ .

SIMPLE has a fixed harvest index and therefore ignores the high sensitivity of grain number to shading around flowering (Fischer, 1985; Andrade et al., 1999). It also ignores the differences in shade tolerance between crop types. In the meta-analysis of Laub et al. (2022), maize and grain legumes lost yield strongly even at low shade levels, whereas C3 cereals and tuber crops initially lost less than proportionally; shading trials also show marked effects of shade on maize development (Ramos-Fuentes et al., 2023). We added a stage-dependent penalty. During a critical window

( $0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$ ), the harvest index is multiplied by  $1 - h\bar{d}$ , where  $\bar{d}$  is the mean relative radiation deficit  $1 - s$  in that window. The sensitivity  $h$  is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al. (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice 0.5, for which no specific curve was available) and lowest for potato (0.2). These values are assumptions that reproduce this ranking; they are not calibrated estimates. Section 3.6 shows the resulting shade-response curves. Because RUE is linear, SIMPLE does not credit the higher light-use efficiency of shaded canopies under more diffuse light. The penalty, conversely, makes shading around flowering more damaging than proportional. The net effect is closer to the field evidence than either assumption alone.

The thermal-time requirement to maturity was calibrated for every site as the mean thermal time between the GGCM sowing and maturity dates under the 2001–2020 climate, and the canopy parameters were scaled accordingly. The cultivar is thus adapted to the present climate, and warming shortens the season. Growth stops smoothly once the thermal-time requirement is met. Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site is represented by a rainfed and an irrigated sub-field in proportion to its SPAM irrigated fraction. Each sub-field carries its own array, so both yield and electricity are area-weighted over the two sub-fields.

### 2.3. SALS: simulation-trained adaptive light sharing

SALS learns a *control network*  $u_t = \pi_{\omega}(\mathbf{x}_t; g)$  (Algorithm 1). It is a multilayer perceptron with two hidden layers of 64 units and a bounded-logit output,  $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$ , which keeps gradients non-zero at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs are the information available on day  $t$ :

- the crop and soil state: relative thermal time  $\text{TT}/T_{\text{sum}}$ , radiation interception  $f_{\text{Solar}}$ , relative soil water and the previous day’s ARID;
- a *forecast* of the day’s maximum temperature (relative to  $T_{\text{heat}}$  and  $T_{\text{opt}}$ ), irradiance and  $\text{ET}_0$ ;
- static information: the irrigation status of the sub-field, the array’s GCR  $g$ , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of  $2^\circ\text{C}$  for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for  $\text{ET}_0$ .

292 The physics always uses the true weather. The same forecast errors are  
 293 applied to the rule-based controllers. The controller is conditioned on  $g$ .  
 294 During training, each season is simulated with a GCR close to the density  
 295 that is sustainable at its training site. This is the largest value on a coarse  
 296 grid  $(0.05, 0.10, \dots, 0.50)$  at which seasonal sharing keeps the mean retention  
 297 of the site’s training seasons at or above  $\rho$ , plus a uniform perturbation in  
 298  $[-0.05, +0.10]$  truncated to  $[0.05, 0.50]$ . The network therefore learns to con-  
 299 trol the densities that design selection will later pick. Outside the cropping  
 300 season the trackers backtrack for maximum electricity.

301 For site-season  $i$  with open-field yield  $Y_i^{\text{open}}$ , let  $r_i = Y_i/Y_i^{\text{open}}$  be the  
 302 yield retention and  $e_i = E_i/E_{\text{PV},i}$  the electricity relative to a conventional  
 303 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[ -e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

304 which combines an exact (hinge) and a quadratic penalty for the food-security  
 305 floor  $\rho$ . Because  $r_i$  and  $e_i$  are outputs of the differentiable simulator, the  
 306 exact gradient of Eq. (3) is obtained by back-propagation through the 365  
 307 daily steps of every season (back-propagation through time). This is the key  
 308 difference from reinforcement learning with black-box simulators (Gautron  
 309 et al., 2022), which estimates gradients from sampled returns.

310 Each minibatch contains 512 site-seasons, each simulated as a rainfed and  
 311 an irrigated sub-field, drawn from one of the four climates in turn. Seasons  
 312 with a failed open-field crop ( $Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$ ) are excluded from training  
 313 and evaluation. The network is fine-tuned through the simulator with Adam  
 314 (Kingma and Ba, 2015) (learning rate  $10^{-3}$ , cosine decay, gradient-norm  
 315 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in  
 316 a numerically stable form, so that gradients remain finite even under extreme  
 317 heat.

318 *Initialisation.* Trained from random weights, the controller converged to  
 319 sharing light throughout the cropping window ( $u > 0.99$  on all days), i.e. to  
 320 the seasonal-sharing rule, because the gradient of electricity with respect to  $u$   
 321 is small and of the same sign for most days of a season. We therefore initialise  
 322 the network by behaviour cloning of the phenology rule of Section 2.4: binary  
 323 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about  
 324 1.7 million day-states visited by the rule in the training seasons, with 1500  
 325 Adam steps at a learning rate of  $3 \times 10^{-3}$ . Fine-tuning then moves the  
 326 policy away from this initialisation wherever doing so improves the objective.

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**Algorithm 1** SALS training

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**Require:** training site-seasons in four climates; floor  $\rho$ ; penalties  $\mu_1, \mu_2$

- 1: initialise  $\pi_\omega$  by behaviour cloning of the phenology rule
- 2: **for** step =  $1, \dots, N$  **do**
- 3:     sample a climate and a minibatch  $\mathcal{B}$  of site-seasons (rainfed and irrigated sub-fields)
- 4:     set  $g_i$  to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5:     **for** day  $t = 1, \dots, 365$  **do**
- 6:          $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$  from state and noisy forecast if the crop is in the field, else 0
- 7:         hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature,  $ET_0$
- 8:         advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9:          $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$  (area-weighted); loss  $\mathcal{L}$  (Eq. 3)
- 10:     back-propagate through the season; Adam update of  $\omega$

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327 Consequently, the gain of SALS over the phenology rule measures what fine-  
328 tuning through the simulator adds to that rule, not what can be learned  
329 without it.

330 *2.4. Deployable design selection and baselines*

331 A real array has one fixed density, which must be chosen before it is built.  
332 Every method, SALS included, therefore uses the same *history-based design*  
333 *rule*. For a site, the GCR is the largest value on the grid  $\{0.05, 0.075, \dots, 0.50\}$   
334 for which the method’s mean yield retention over the site’s 2001–2014 sea-  
335 sons is at least  $\rho = 0.9$ . If no value qualifies, no array is built and the site  
336 remains open field. A stricter *chance-constrained* variant requires  $r \geq 0.9$  in  
337 at least 80% of the historical seasons. The historical seasons of the test sites  
338 are never used for training, and the weather of the evaluation years is never  
339 used for design. We compare four controllers:

- 340     • *AV static*: backtracking trackers without light sharing ( $u = 0$ );
- 341     • *Seasonal sharing*: the light-sharing rotation ( $u = 1$ ) throughout the  
342         cropping season;

- *Stress rule*: an expert rule that shares light ( $u = 1$ ) during the main growth phase ( $0.1 < TT/T_{\text{sum}} < 0.9$ ) except on days with a heat-stress forecast ( $T_{\text{max}} \geq T_{\text{heat}}$ ) or drought ( $\text{ARID} > 0.5$ ), when the modules track the sun and shade the crop;
- *SALS*: the learned controller.

The open field and a conventional PV plant without crops ( $\text{LER} = 1$  each) are the references. Because the design rule is identical for all controllers, differences between methods reflect the quality of control and the array density that each controller can sustain under the food-security floor. We also compare the controllers at identical GCR over the whole grid (matched-density frontiers).

## 2.5. A shade-response model learned from field data

The simulator needs a crop response to shade that is supported by observations. We learned one from the open data of the meta-analysis of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as  $y(s) = \exp(-\beta s^\gamma)$  with Student- $t$  errors ( $\nu = 4$ ; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss  $y = 1 - s$ ; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of  $\log \beta$  and  $\log \gamma$  towards a common value, scale 0.5); partial pooling with covariates on  $\log \beta$  (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

## 2.6. Shade-response uncertainty: gradient calibration, robust controllers and design

The crop’s response to shade is the least certain part of the simulator, so we treat it as an unknown with a distribution. Two parameters per crop

378 enter the simulator: a radiation-use compensation  $c$  (growth is multiplied by  
 379  $1 + c(1 - s)$ ,  $s$  being the share of open-field irradiance that reaches the crop)  
 380 and a multiplier  $h$  on the flowering-window harvest-index penalty. Because  
 381 the simulator is differentiable, both can be fitted by gradient descent through  
 382 whole simulated seasons to published yield–shading observations. The ob-  
 383 servations are the relative yields at 20% and 40% shading of the data-driven  
 384 curves of Section 2.5 for the five crops (median, with an error equal to the  
 385 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice  
 386 takes the curve of C3 cereals, which contains no rice), the field trials on rice,  
 387 wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-  
 388 Fuentes et al., 2023) (each season of each trial as a separate observation, error  
 389 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes  
 390 et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses  
 391 the same procedure with a target error of 0.2 for the learned curves, which  
 392 is the between-study spread of a new study (Section 3.7); the spread of  
 393 the resulting members is smaller (standard deviation of relative yield across  
 394 members 0.04–0.09 at 20% shading and 0.05–0.12 at 40% shading, against  
 395 0.02–0.04 and 0.03–0.06 for E2), because the fit also weights the field trials  
 396 and the priors. An earlier ensemble, E1, was fitted in the same way to curves  
 397 read by hand from the figures of Laub et al. (2022) and was used to train  
 398 the robust and adaptive controllers below. Priors are  $c \sim \mathcal{N}(0.4, 0.5^2)$  and  
 399  $h \sim \mathcal{N}(1, 0.5^2)$ , and the parameters are clamped to  $[0, 2.5]$  ( $c$ ) and  $[0, 2]$  ( $h$ )  
 400 after every step, which is why a response milder than the data cannot be  
 401 lowered further. We fitted 12 ensemble members by randomised maximum a  
 402 posteriori estimation: every member sees the observations perturbed by their  
 403 errors and a prior draw perturbed by its width, and is optimised with Adam  
 404 (40 steps, with the loss still falling slightly at the last step; 40 training-block  
 405 seasons per crop and member; shading of the static arrays cached). The  
 406 ensemble is a device for generating plausible responses and not a validated  
 407 posterior: it has no coverage check, and for some parameters (the harvest-  
 408 index multiplier of wheat and potato) its spread equals that of the prior.  
 409 The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12),  
 410 chosen in advance. The ensemble is used in three ways. All training and  
 411 evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes  
 412 and the calibration of one ensemble about 7 minutes.

413 *Robust and adaptive controllers.* SALS-robust has the architecture and  
 414 training of SALS (Section 2.3), but every training season is simulated with the  
 415 response of a randomly drawn ensemble member that the controller does not

416 observe, and the density jitter is widened to  $[-0.05, +0.25]$  because feasible  
 417 densities differ between members. SALS-adaptive additionally observes the  
 418 realised biomass of the crop under the modules relative to an open reference  
 419 plot, a quantity that can be measured on the farm by remote sensing or by  
 420 a reference plot. A feedback rule that shares light whenever this ratio falls  
 421 below a target serves as the hand-written counterpart; its target (1.0) was  
 422 chosen on training-block seasons under the ensemble.

423 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the  
 424 largest GCR whose historical retention meets the floor. Under shade-response  
 425 uncertainty we choose the largest GCR that meets the floor in at least 75%  
 426 of five ensemble members, and compare it with a design under the ensemble  
 427 mean and with an oracle design under the true response. Evaluation of this  
 428 rule is leave-one-out: the truth is the held-out member and the design uses  
 429 the other four.

430 *Stress tests.* Controllers and rules are also evaluated under four fixed  
 431 responses, the proportional response, the response calibrated to the meta-  
 432 analysis, the response fitted to the field trials and a harsh response (propor-  
 433 tional growth and a 1.5 times larger harvest-index penalty). The proportional  
 434 and harsh responses lie outside the range of the ensemble.

## 435 2.7. Economic analysis

436 To judge whether the electricity gains matter economically, we computed  
 437 an annual net value per hectare for each method relative to the open field:  
 438 electricity revenue minus the annuity and operating cost of the installed ca-  
 439 pacity minus the loss of crop revenue (crop production costs are unchanged).  
 440 The installed capacity is 2.1 g MWp per hectare (modules cover a share  $g$  of  
 441 the land at  $0.21 \text{ kWp m}^{-2}$ ). Yield is dry matter, valued at assumed prices.  
 442 A conventional ground-mounted plant (GCR 0.4) that replaces the crop is  
 443 the comparator. All parameters (Table 15) are assumptions chosen to be  
 444 representative, not data, and their ranges drive the conclusions; we therefore  
 445 report break-even electricity prices and one-at-a-time sensitivities, and we do  
 446 not monetise food security, which market crop prices do not capture.

## 447 2.8. Evaluation protocol, statistics and upscaling

448 Sites were split spatially:  $10^\circ \times 10^\circ$  blocks were randomly assigned to train-  
 449 ing (70%) or test (30%), so that no test site has a training site in the same  
 450 block. Training uses the 2001–2014 seasons of the training sites. Design  
 451 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the

2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- the mean electricity ( $\text{MWh ha}^{-1} \text{yr}^{-1}$ ) and the relative electricity  $e$ ;
- the mean yield retention  $r$  and the production-weighted retention  $\sum Y / \sum Y^{\text{open}}$ ;
- season-level compliance (share of seasons with  $r \geq \rho$ ) and site-level compliance (share of sites whose mean test-season retention is  $\geq \rho$ );
- the lower tail (10th percentile of  $r$ , share of seasons with  $r < 0.8$ );
- the land-equivalent ratio  $\text{LER} = r + e$  (Dupraz et al., 2011).

Two aggregations are used and labelled in the text: tables report means over all test seasons of a sample (season-pooled), whereas statistical comparisons and intervals use the crop–site pair as the unit of analysis (mean of the pair’s seasons, then mean over pairs), so the same contrast can differ slightly between the two (for example 0.034 and 0.038 for SALS minus the phenology rule on the main split). Samples are also labelled: the main analysis uses the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses pooled over three splits use the 76 distinct pairs of every third held-out pair and a grid with step 0.05, and analyses of alternative shade responses on one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

### 3. Results

#### 3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out crop–site pairs of the main split (95 pairs are held out; pairs without a season whose open-field

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity  $h$  added in this study.  $T_{\text{sum}}$ ,  $I_{50A}$  and  $I_{50B}$  are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in  $^{\circ}\text{C}$ , thermal times in  $^{\circ}\text{C d}$ , RUE in  $\text{gMJ}^{-1}$ ;  $S_{\text{CO}_2}$  is the relative RUE increase per 100 ppm above 350 ppm.

| Crop (cultivar)       | $h$ | $T_{\text{sum}}$ | HI   | $I_{50A}$ | $I_{50B}$ | $T_{\text{base}}$ | $T_{\text{opt}}$ | RUE  | $I_{50 \text{ max } H}$ | $I_{50 \text{ max } W}$ | $T_{\text{heat}}$ | $T_{\text{ext}}$ | $S_{\text{CO}_2}$ | $S_{\text{water}}$ |
|-----------------------|-----|------------------|------|-----------|-----------|-------------------|------------------|------|-------------------------|-------------------------|-------------------|------------------|-------------------|--------------------|
| Wheat (Batten)        | 0.4 | 2150             | 0.34 | 280       | 50        | 0                 | 15               | 1.24 | 100                     | 25                      | 34                | 45               | 0.08              | 0.4                |
| Rice (IR72)           | 0.5 | 2300             | 0.47 | 850       | 200       | 9                 | 26               | 1.24 | 100                     | 10                      | 34                | 50               | 0.08              | 1.0                |
| Maize (McCurdy 84aa)  | 0.6 | 2050             | 0.50 | 500       | 50        | 8                 | 28               | 2.10 | 100                     | 12                      | 34                | 50               | 0.01              | 1.2                |
| Soybean (Williams 82) | 0.5 | 2350             | 0.40 | 600       | 200       | 6                 | 27               | 0.86 | 120                     | 20                      | 36                | 50               | 0.07              | 0.9                |
| Potato (Sebago)       | 0.2 | 2400             | 0.85 | 500       | 350       | 4                 | 22               | 1.30 | 50                      | 30                      | 34                | 45               | 0.10              | 0.4                |

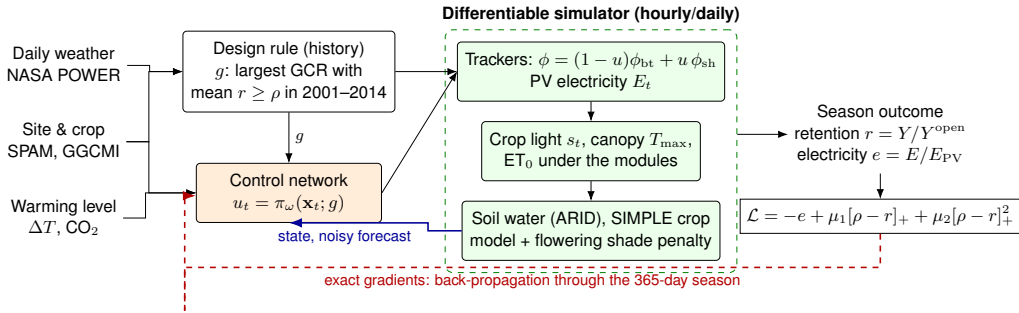


Figure 1: Simulator and controller. The array density  $g$  of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density  $g$  and a noisy forecast to the light-sharing level  $u_t$ . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

yield is at least  $0.2 \text{ t ha}^{-1}$  in all four climates are excluded) (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

**Static arrays are the weakest.** Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce  $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ , 16% of a conventional PV plant per hectare (LER 1.07). Rotating the modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and  $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$  (LER 1.19). A rule that shares light only while the crop is growing (phenology rule) reaches

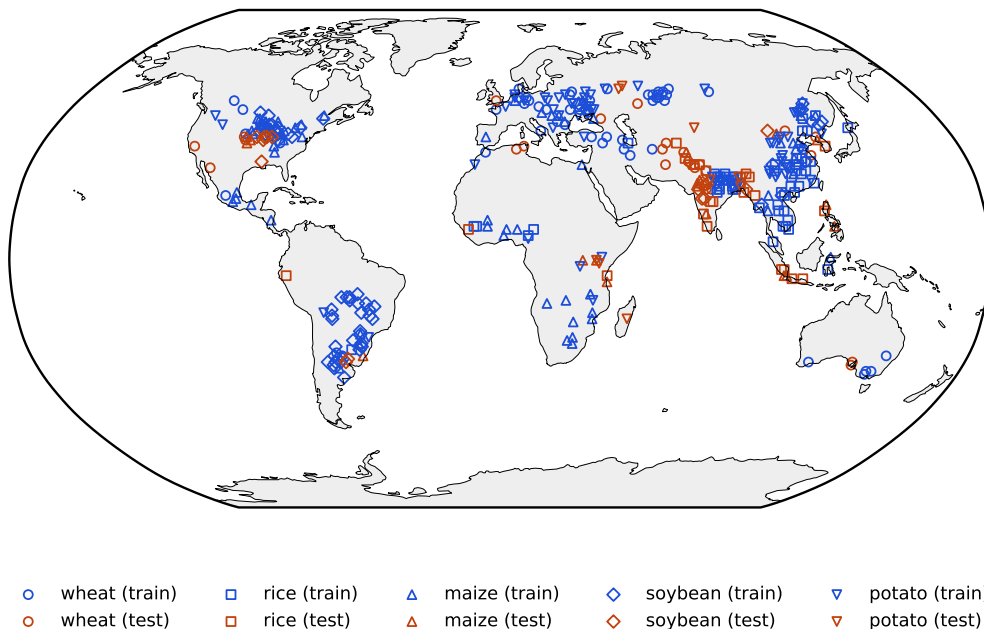


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

496 465 MWh ha<sup>-1</sup> yr<sup>-1</sup> (LER 1.24). An expert rule that additionally shades the  
 497 crop on forecast heat or drought days (stress rule) is worse than the phenol-  
 498 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at  
 499 times when the floor is most binding.

500 **The learned controller is best by a modest, consistent margin.**  
 501 SALS sustains a density of 17% (similar to the phenology rule), produces  
 502 510 MWh ha<sup>-1</sup> yr<sup>-1</sup> and reaches an LER of 1.270 at the same retention. Com-  
 503 pared with the phenology rule this is +0.038 in LER (95% CI over sites  
 504 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-  
 505 ble A.18). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35  
 506 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows  
 507 the same picture at matched densities: the learned policy lies on or slightly  
 508 above the phenology rule over the whole density range, both lie well above  
 509 seasonal sharing and the stress rule, and static arrays are far below.

510 **What the floor does and does not guarantee.** Enforcing the floor

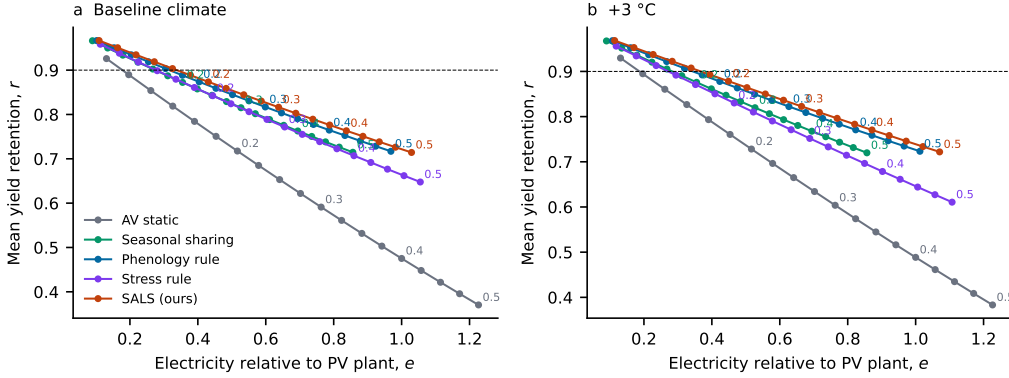


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have  $r \geq 0.9$  (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have  $r < 0.8$  (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of  $r$  is 0.883, against 0.887, 0.886 and 0.903. Requiring  $r \geq 0.9$  in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

### 3.2. Robustness

**Seeds and splits.** Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-pooled: 0.034 and 0.032; the site-level paired contrasts of Table A.18 are 0.038 and 0.034). In every split SALS ranks first, the phenology rule second and static arrays last.

**Canopy cooling.** The canopy-cooling coefficient  $\kappa$  is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with  $\kappa = 0, 1$  and 2 °C. Electricity changes with  $\kappa$  because cooler canopies tolerate denser arrays (SALS: 475, 510 and

533 549 MWh ha<sup>-1</sup> yr<sup>-1</sup> under the baseline climate), but the retention is held at  
 534 0.905 by the design rule and the ranking is unchanged. SALS exceeds the  
 535 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate  
 536 (0.022, 0.032 and 0.051 at +3 °C).

537 **Which ingredients matter.** Two ablations test the design of SALS  
 538 at the same training budget. Masking the stress inputs (soil water, ARID,  
 539 and the maximum temperature, irradiance and ET<sub>0</sub> forecasts) leaves the  
 540 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297  
 541 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The  
 542 improvement over the rules therefore does not depend on stress-aware inputs  
 543 or on training in warmed climates.

### 544 *3.3. Floor conditioning, chance-constrained training and forecast errors*

545 **One network for all floors.** Regulations and farmers set different  
 546 floors. The network of the main experiments was trained at  $\rho = 0.9$  only.  
 547 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical  
 548 design step for each floor (Table 3). Its LER exceeds that of the phenology  
 549 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,  
 550 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without  
 551 retraining. A network that received the floor as an additional input and was  
 552 trained over all four floors did not improve on it (differences in LER  $\leq 0.001$ ).  
 553 Because the design step already adapts the density to the floor, conditioning  
 554 the controller on the floor is not needed.

555 **Layout, rotation limit and harvest-index assumptions.** We checked  
 556 three assumptions that could drive the ranking (Table 4; 21 pairs of the main  
 557 split, one training seed). The rotation limit of the trackers sets how much  
 558 light sharing can deliver: with a limit of 45° the LER of the phenology rule  
 559 is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at  
 560 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking  
 561 is unchanged. Fixed non-tracking arrays held flat or rotated by 30° about  
 562 the north–south axis (facing east, not the equator; equator-facing tilts are  
 563 not modelled) do no better than static trackers (LER 1.048 and 1.043 against  
 564 1.053; differences of 0.005 are within the noise of 21 pairs and one training  
 565 seed), so the static tracker is not a weak baseline relative to them; verti-  
 566 cal bifacial layouts are not modelled. The ranking of the rules also does not  
 567 depend on the assumed harvest-index penalty. The window test is weakly in-  
 568 formative about the timing of the phenology rule, which shares light through  
 569 almost the whole growing period, so a shifted window cannot move it much;

the informative case is the removal of the penalty. Moving the critical window to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement changes the LER of the phenology rule by at most 0.001, and removing the penalty altogether raises all light-sharing LERs (phenology rule 1.328, SALS 1.353) without changing the order: the phenology rule exceeds seasonal sharing by 0.03 to 0.05 in every case (0.035 with the early window, 0.034 with the late window, 0.053 without penalty), because it produces electricity before canopy development and after maturity, when the crop does not use the light, and not because of the assumed window. The margin of SALS over the written rule changes with these assumptions (0.005 to 0.025 LER), and its margin over the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS equals the tuned rule).

**A floor of 66%.** The German pre-standard DIN SPEC 91434 requires 66% of the reference yield (DIN, 2021). Repeating the design step with  $\rho = 0.66$  (baseline climate, main split, same network trained at 0.9) the ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and SALS 1.71. At this floor the density is limited by the cap and by the mean retention of 0.73 reached at the cap, not by the floor, so the comparison of controllers says little about control; we do not interpret it further.

**Chance-constrained training.** We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

**Forecast errors.** We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The stress rule, which depends on the forecast of maximum temperature, degrades with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and its LER from 1.432 to 1.418 as the errors triple.

**A tuned rule.** The phenology rule above was written, not tuned,

608 whereas SALS is trained. We therefore tuned the window of the rule on  
 609 the training-block seasons with the same objective (electricity at the density  
 610 where mean retention equals the floor); the best window starts at 0.15 in-  
 611 stead of 0.05 of the thermal-time requirement (Table 5). Pooled over three  
 612 splits and 76 distinct crop–site pairs, the tuned rule raises LER over the orig-  
 613 inal rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS  
 614 exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at  
 615 equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the  
 616 rule thus accounts for about 60% of the margin of SALS over the written  
 617 rule. A five-parameter ramp-plateau schedule optimised by CMA-ES on the  
 618 same training seasons performs like the tuned window (LER 1.238 against  
 619 1.233, difference +0.005, interval  $-0.001$  to  $+0.010$ ), and SALS exceeds it  
 620 by 0.013 (0.007 to 0.019); the optimum is an almost binary window (start  
 621 0.17, end 0.93, level 0.99), so more flexible schedules of this family do not  
 622 close the gap. The remaining margin of SALS is small but not zero; it is not  
 623 explained by the tuning of a window or by a smooth ramp, and we did not  
 624 test richer rule families (for example schedules that depend on soil water).  
 625 SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

### 626 3.4. *What the learned policy does*

627 Averaged over the rainfed test seasons of the main split at a density of 0.3,  
 628 the light-sharing level of SALS is a smooth function of relative thermal time.  
 629 It is close to zero in the first 10% of the thermal time required for maturity,  
 630 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and  
 631 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers  
 632 the window in which our crop model penalises shading through the harvest  
 633 index. The ramps at both ends let the modules produce electricity when  
 634 additional light is worth least to the crop. Fig. 4 shows two example seasons.  
 635 The phenology rule, which is also the initialisation of the network, shares  
 636 light during the whole growing period. Fine-tuning through the simulator  
 637 found that sharing can be reduced at the edges of the season: SALS gains its  
 638 advantage by yielding light when it matters and collecting electricity when  
 639 it matters less.

640 The policy also shares less light on days with a forecast heat stress: the  
 641 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ( $+3^\circ\text{C}$ )  
 642 on other days, and 58–63% of sites show a lower level on heat days. This  
 643 cannot be read as a stress response, however. Heat days occur late in the  
 644 season, when the phenological ramp is already falling, and a policy without

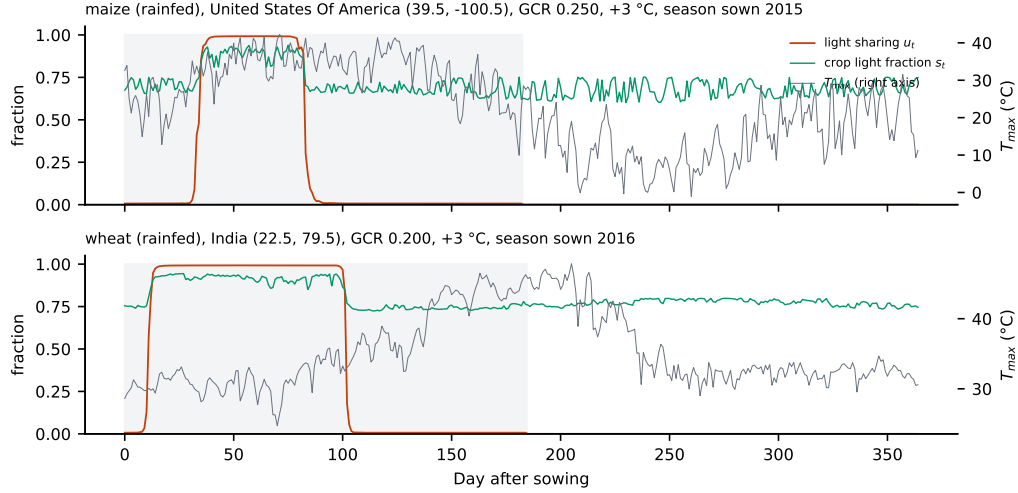


Figure 4: Learned daily control for two test seasons under  $+3^{\circ}\text{C}$ , at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level  $u_t$ ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).  
 The ablation above confirms that stress information does not improve the  
 result.

### 3.5. Crops and regions

The benefit of light sharing differs between crops (baseline climate, SALS). Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and 577 MWh ha<sup>-1</sup> yr<sup>-1</sup>; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20 for both). The stress rule has a lower LER for every crop (1.26 for soybean, 1.14–1.21 for the others).

The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose more than low-yielding ones. Regionally, Latin America and Oceania (4 sites) combine the highest density (GCR 0.41) and electricity (1090 MWh ha<sup>-1</sup> yr<sup>-1</sup>) with a production-weighted retention of only 0.862. Africa and the Middle East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39 pairs

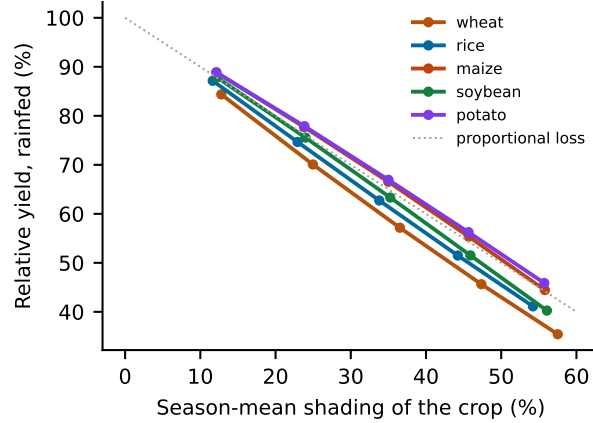


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

without intervals and should be read as indications. A regulatory floor enforced per site would therefore need a region- and crop-specific design step, not a global one.

### 3.6. Model behaviour and validation

**PV output.** The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with  $r = 0.971$ , a mean bias of  $-4.9\%$  and a mean absolute deviation of  $5.5\%$  (Section 2.2).

**Shade response.** Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

**Comparison with field trials.** We fitted one radiation-use compen-

683 sation coefficient per crop to the yield–shading curves of Laub et al. (2022)  
 684 using training-block sites only (Methods, calibration script in the repository).  
 685 The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the  
 686 proportional model already lies within about 0.02 of the C3-cereal and tu-  
 687 ber curves, and it cannot lower the maize and legume responses below the  
 688 proportional case (root-mean-square error 0.16 for maize and 0.08 for soy-  
 689 bean). Against independent field trials (Table 6) the model underestimates  
 690 retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and  
 691 potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al.  
 692 (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91  
 693 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was  
 694 not affected by shade. The model is therefore conservative for food reten-  
 695 tion with respect to these trials, except for the stronger maize losses of the  
 696 meta-analysis. Table 7 shows the consequence for the design results: with  
 697 the calibrated response the allowed density and the LER of every method  
 698 rise (phenology rule 1.28 against 1.23), and with a response fitted to the  
 699 field trials the LER of the phenology rule is 1.62. The phenology rule is the  
 700 best method in all three cases, and light sharing adds LER +0.17, +0.19 and  
 701 +0.30 over static arrays (conservative, calibrated, field-fitted); only the order  
 702 of the seasonal and stress rules changes. The absolute gains for rice, wheat  
 703 and potato are thus conservative with respect to these trials; for maize and  
 704 soybean the meta-analysis points the other way (Section 3.7).

### 705 *3.7. A shade-response model learned from field data*

706 Cross-validation by study shows what the data can and cannot support  
 707 (Tables 8 and 9). Over all 339 shaded observations the root-mean-square  
 708 error of a held-out study falls from 0.43 for proportional loss  $y = 1 - s$  to 0.37  
 709 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes  
 710 from crop types that the simulator does not represent (berries, forages, fruits  
 711 and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses,  
 712 the learned curves are not better than proportional loss: the error is 0.222  
 713 against 0.220 pooled (difference +0.002, 95% interval  $-0.023$  to  $+0.030$ ),  
 714 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes  
 715 and tubers. Partial pooling is as good as separate curves (0.312 against 0.313)  
 716 and the covariates add almost nothing (0.311). Gradient-boosted trees on the  
 717 same inputs are worse (0.321) than the parametric model. The error of 0.22  
 718 for the four crop types is large: yields of a new study differ from the learned  
 719 curve by about 20 percentage points, which is the between-study spread

720 that a design under uncertainty has to cover. Crop type is the information  
721 that transfers between studies, and none of the covariate effects on  $\log \beta$  is  
722 clearly different from zero (irradiation  $-0.10$ , 90% interval  $-0.32$  to  $0.09$ ;  
723 subtropical zone  $-0.37$ ,  $-0.73$  to  $0.07$ ; intercropping  $0.16$ ,  $-0.48$  to  $0.45$ ).  
724 The estimated effect of solar-panel shade is not interpretable, because all 46  
725 panel observations are of leafy vegetables, mainly lettuce, and the effect is  
726 confounded with crop type.

727 The learned curves show a clear ordering of crop types (Table 10). At  
728 20% shading C3 cereals and tubers lose less than proportionally (relative  
729 yield  $0.93$ ), whereas maize ( $0.73$ ) and grain legumes ( $0.71$ ) lose more; at 40%  
730 shading the values are  $0.72$  and  $0.71$  against  $0.42$  and  $0.40$ . Intervals are wide  
731 for maize ( $0.27$ – $0.59$  at 40%) and for tubers, which rest on two studies. The  
732 simulator’s responses are close to the data for cereals and potato but, under  
733 both calibrated ensembles, too mild for maize and soybean at 40% shading  
734 ( $0.75$  and  $0.48$  for E2 against  $0.42$  and  $0.40$  in the data). The calibration  
735 cannot lower the maize response further, because the simulated shading also  
736 relieves heat and drought stress and the compensation and harvest-index  
737 parameters are bounded below at zero. The field trials of maize are also  
738 milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest  
739 index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6%  
740 above the control under a low-density stilt-mounted array in one season at  
741 one site. For maize and soybean the direction of the bias is uncertain, and  
742 the gains reported in this paper are possibly overestimated.

### 743 3.8. Learning under shade-response uncertainty

744 **Calibration.** Gradient descent through the simulator fitted the five crops  
745 to the data-driven curves and the field trials (ensemble E2, Table 11). The  
746 data identify two features: the harvest index of maize is hardly affected by  
747 shade (multiplier  $0.16 \pm 0.13$ ,  $c \approx 0.14$ ), as observed by Ramos-Fuentes et al.  
748 (2023), and soybean is more shade sensitive than the cereals (compensation  
749  $0.05$ , harvest-index multiplier  $1.66$ ), as in Laub et al. (2022). Wheat has the  
750 largest compensation ( $0.91 \pm 0.31$ ) and the largest uncertainty; the compen-  
751 sation is an effective parameter that absorbs diffuse-light and light-saturation  
752 effects that the simulator does not model explicitly, not a physical value.

753 **Learned controllers.** Controllers were evaluated under four fixed re-  
754 sponses on 21 crop–site pairs of split 0 with one training seed, so the intervals  
755 below, which are cluster bootstraps over pairs of per-pair mean LER differ-  
756 ences, do not include seed or split variation (Table 12 gives season means,

757 which differ slightly). Under every response SALS, trained on the propor-  
 758 tional response, is better than the written phenology rule: by +0.019 (0.007  
 759 to 0.039), +0.024 (0.009 to 0.045), +0.038 (0.017 to 0.063) and +0.015 (0.000  
 760 to 0.036) LER under the proportional, calibrated, field-fitted and harsh re-  
 761 sponses. Training on the narrow ensemble E1 did not change this: SALS-  
 762 robust differs from SALS by  $-0.005$  to  $+0.003$  (intervals contain zero except  
 763 for the proportional response, 0.000 to 0.003), and SALS-adaptive, which  
 764 sees the measured biomass ratio, by  $-0.010$  to  $+0.007$  (all intervals contain  
 765 zero). Training on the wider ensembles did not help either: SALS-robust on  
 766 E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017 and  
 767 0.017 LER under the proportional response, 0.017, 0.021 and 0.020 under the  
 768 calibrated response, 0.004, 0.005 and 0.011 under the field-fitted response (in-  
 769 tervals contain zero) and 0.008, 0.014 and 0.017 under the harsh response;  
 770 the intervals exclude zero in 6 of the 9 comparisons for the proportional, cal-  
 771 ibrated and harsh responses. The E3 robust controller is not distinguishable  
 772 from the written rule ( $+0.001$  to  $+0.003$ , intervals contain zero) except un-  
 773 der the field-fitted response ( $+0.032$ , 0.001 to 0.067). A tuned window and a  
 774 CMA-ES ramp schedule, also evaluated under the four responses (Table 12),  
 775 are 0.005 to 0.018 LER below SALS (intervals of the SALS contrast:  $+0.004$   
 776 to  $+0.016$  against the tuned window under the proportional response, and  
 777 containing zero for the ramp under the calibrated and field-fitted responses  
 778 and for the window under the harsh response), and the ensemble-trained  
 779 controllers are at or below the tuned window under all responses except the  
 780 field-fitted one. With one seed and 21 pairs these controllers therefore did not  
 781 retain the margin of SALS within our training budget; a controller that must  
 782 perform across plausible responses is less specialised, but a larger budget or  
 783 more seeds could change this. The feedback rule is worse than the written  
 784 rule by 0.026–0.058 LER under every response (intervals exclude zero). A  
 785 plausible reason is that the biomass ratio does not reveal the harvest-index  
 786 loss from shading around flowering, but we did not test this explanation.

787 **Design under uncertainty.** The larger risk is in the design step (Ta-  
 788 bles 13 and 14, Fig. 6). The designs below are evaluated by leaving one  
 789 ensemble member out as the truth and using the other four, so that the  
 790 truth never enters the design. Pooled over the three spatial splits (76 dis-  
 791 tinct crop–site pairs), a density chosen under the mean response of E2 leaves  
 792 70% of the pairs at the floor when the true response is another plausible one,  
 793 against 87% for an oracle design (E3: 71% and 85%). Choosing the density  
 794 so that the floor is met in at least 75% of the other members raises compli-

795 ance by +7.9 points under E2 (95% cluster-bootstrap interval 4.5 to 11.6)  
 796 and +7.9 under E3 (3.9 to 12.4), at a cost of 0.031 and 0.025 in LER. Raising  
 797 the floor used in the design from 0.90 to 0.92 under the mean response does  
 798 more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and +14.2 points  
 799 (9.7 to 19.2) at 0.056 under E3, and the pairs below the floor fall short by less  
 800 (mean shortfall 0.0022 against 0.0045 for the quantile rule under E2). The  
 801 two kinds of rule lie on one frontier (Fig. `effig:designfrontier`). Interpolating  
 802 the margin frontier (floors 0.90 to 0.96) at the compliance of the ensemble  
 803 rules, the LER of the margin minus that of the rule is +0.001 (95% inter-  
 804 val  $-0.009$  to  $0.012$ ) for the rule with 75% of the other members and +0.003  
 805 ( $-0.016$  to  $0.021$ ) for the rule with all of them under E2, and  $-0.006$  ( $-0.019$   
 806 to  $0.005$ ) and +0.024 ( $-0.011$  to  $0.058$ ) under E3, so a simple margin and  
 807 the ensemble rules are not distinguishable on this sample. The ensemble is  
 808 thus useful for sizing the margin (0.02 to 0.04 above the floor for 85–97%  
 809 of the pairs), not for replacing it. None of these rules is safe outside the  
 810 ensemble (Table 14): under the proportional, calibrated and harsh responses  
 811 the mean-response design leaves 7–17% of the pairs at the floor, a floor of  
 812 0.92 or the ensemble rules 12–33%, a floor of 0.94 29–47% and a floor of 0.96  
 813 51–78%, at a cost of about 0.2 LER, whereas the field-fitted response, which  
 814 lies inside the range spanned by the ensembles, is handled well by every rule  
 815 (80–99%) except the mean-response design (68–79%). The guarantee there-  
 816 fore holds only within the range of responses that the ensemble spans, and a  
 817 response harsher than the data supported by the ensemble breaks it.

### 818 3.9. *Economics of light sharing*

819 All figures in this section follow from the assumed parameters of Table 15  
 820 and should be read as a sensitivity analysis, not a forecast. The economics  
 821 are dominated by the capacity installed, not by the crop (Table 16): the  
 822 annuity of the modules is 11–32 kUSD per hectare and year, whereas the  
 823 crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per  
 824 installed watt. With the densities designed for the floor, static arrays earn  
 825 +0.3 kUSD ha<sup>-1</sup> yr<sup>-1</sup> at 60 USD MWh<sup>-1</sup> and break even at 59, whereas all  
 826 light-sharing methods lose money at that price ( $-3$  to  $-10$  kUSD) and break  
 827 even at 70 (stress rule), 73 (SALS), 77 (tuned phenology rule), 80 (written  
 828 rule) and 92 (seasonal sharing) USD MWh<sup>-1</sup>. At 90 USD MWh<sup>-1</sup> static  
 829 arrays (6.3 kUSD) are better than the written (4.0) and tuned (5.5) rules  
 830 and worse than SALS (7.6), which overtakes static arrays above about 85  
 831 USD MWh<sup>-1</sup>. A conventional plant that replaces the crop breaks even at 48

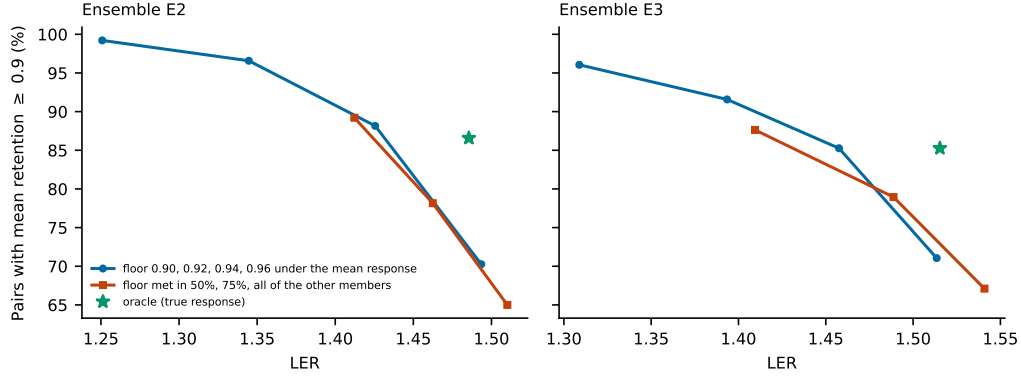


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

USD MWh<sup>-1</sup> and is more profitable than every agrivoltaic option above that price (16 kUSD at 60, 94 kUSD at 120, against -6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh<sup>-1</sup>: plant -23.2, SALS -18.7, static -5.7 kUSD). Per megawatt-hour, the electricity of SALS costs about 73 USD including the crop lost, against 48 USD for the plant, a difference of 25 USD MWh<sup>-1</sup>. A premium of that size would equalise the unit cost but not the profit per hectare, because a hectare of plant produces three times as much electricity (1306 against 438 MWh): relative to the plant, SALS would need a premium of about 49 USD MWh<sup>-1</sup> at a market price of 60 and about 109 at 90 to match its profit per hectare, and only about 13 at 60 (none above 73) to match the open field. The sensitivity is large (Table 17): at 90 USD MWh<sup>-1</sup> the net value of SALS changes from +17 to -8 kUSD between module costs of 0.7 and 1.5 USD Wp<sup>-1</sup>, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30-40 USD MWh<sup>-1</sup>; at 60 14-59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh<sup>-1</sup>, SALS leads from about 80 (4.9

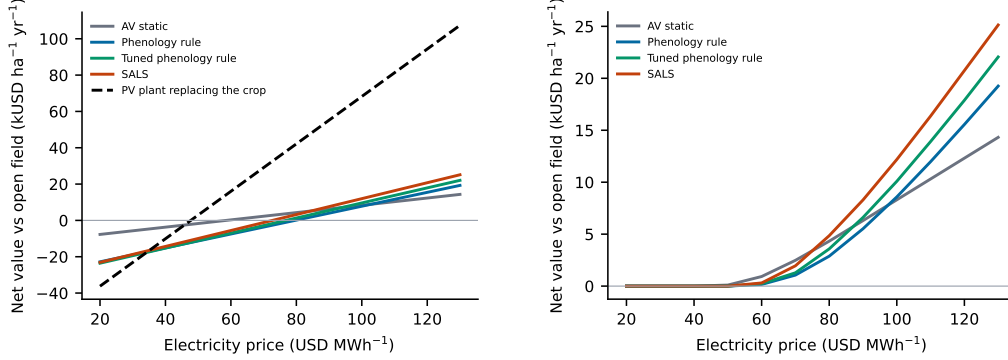


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 15.

853 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), and the  
 854 tuned rule overtakes static arrays only above about 105. The value of control  
 855 is therefore real but depends on a high price: at 90 USD MWh<sup>-1</sup> SALS is  
 856 worth 1.7 kUSD ha<sup>-1</sup> yr<sup>-1</sup> over the tuned rule with profit-optimal densities.

857 **Cost of the safety margin.** At 90 USD MWh<sup>-1</sup>, moving from a density  
 858 designed under the mean response to a floor of 0.92 under that response costs  
 859 1.00 kUSD ha<sup>-1</sup> yr<sup>-1</sup> (17%) under E2 for 18 more points of compliance (70%  
 860 to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about  
 861 56 and 79 USD ha<sup>-1</sup> yr<sup>-1</sup> per point. The ensemble rule that requires all other  
 862 members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points  
 863 (45 and 74 USD per point), so on this measure it is cheaper than the margin  
 864 under E2 and similar under E3. At 60 USD MWh<sup>-1</sup> a margin has a negative  
 865 cost, because every module beyond break-even loses money; the economics  
 866 of the design margin therefore depend on the price level, whereas its effect  
 867 on compliance does not.

### 868 3.10. Agrivoltaics under warming

869 Under warming, the open-field production of the held-out sites (Fig. 8)  
 870 falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soy-  
 871 bean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production  
 872 with light-sharing control follows the open field at about 90% of its level, so  
 873 it inherits the same climate impacts (maize -31%, rice -27% relative to the

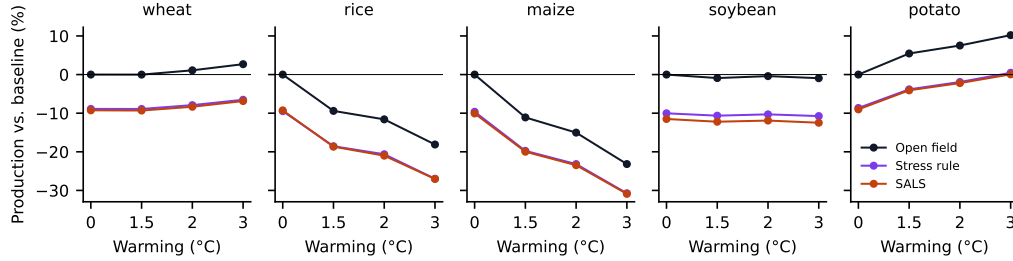


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha<sup>-1</sup> yr<sup>-1</sup>). The margin of SALS over the rules does not grow with warming (Table A.18).

### 3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop–site pairs, in at least one split; 222 are held out in at least one split and three have no valid season), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr<sup>-1</sup> under the baseline climate and 16.0 PWh yr<sup>-1</sup> at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr<sup>-1</sup>, and static arrays 6.4 and 6.9 PWh yr<sup>-1</sup>. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr<sup>-1</sup> and the best 5% 40 PWh yr<sup>-1</sup> with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites

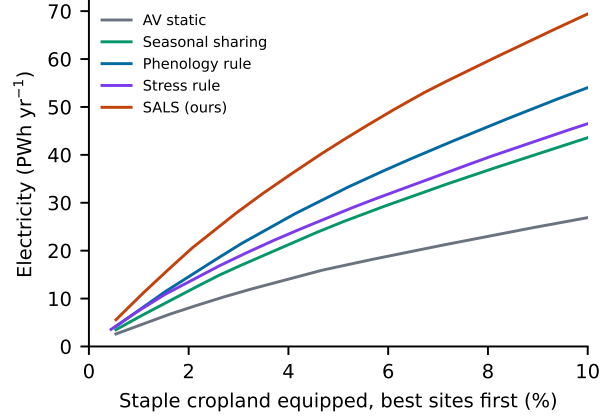


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is probably too mild (Section 3.7), account for 53% of the electricity of the upscaling (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only wheat, rice and potato would give about 7 PWh yr<sup>-1</sup> at the same coverage, so the global figure should be read as an upper estimate for about half of its content. By crop, the LER of SALS under the baseline climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

## 4. Discussion

### 4.1. What the results establish

Four findings stand on consistent evidence. First, sharing light with the crop is the main lever on the electricity that tracker-based agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control allows 2.1 to 2.9 times the density, and the LER rises to 1.18–1.24 for hand-written rules. Second, the best simple rule follows crop phenology. It

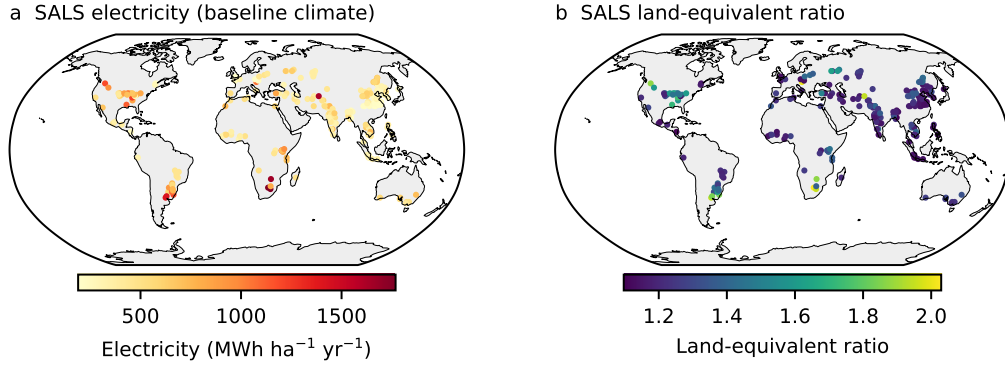


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

919 shares light from the onset of canopy development (5% of the thermal-time  
 920 requirement) until maturity and produces electricity before and after, when  
 921 the crop does not use the light. It needs only the crop’s thermal time, which  
 922 is observable on the farm, and it is insensitive to forecast errors, whereas  
 923 the rule that reacts to heat and drought stress is worse in every climate and  
 924 degrades with forecast error. Third, the qualitative result does not depend  
 925 on the weakest part of the model. The simulated shade response is conser-  
 926 vative against the field trials we could use (Section 3.6); calibrating it raises  
 927 the allowed density, and the phenology rule stays the best method, with a  
 928 gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative,  
 929 calibrated and field-fitted responses. However, the response learned from  
 930 the meta-analysis shows steeper losses for maize and grain legumes than the  
 931 simulator reproduces (Section 3.7), so the absolute gains are conservative  
 932 for wheat, rice and potato and probably overestimates for maize and soy-  
 933 bean (53% of the upscaled electricity). Fourth, a controller learned through  
 934 the differentiable simulator is better than the best rule we could write, by  
 935 0.04 in LER against the written rule (0.018 against a rule whose window  
 936 was tuned on the training seasons, 5.7% more electricity at equal retention),  
 937 reproducibly over training seeds (range 0.002 in LER), three spatial splits,  
 938 three canopy-cooling assumptions and four climates, and at 84–86% of the  
 939 held-out sites. This gain is small compared with the gap between static and  
 940 dynamic control.

941 *4.2. What role does artificial intelligence play?*

942 We tested three ways in which learning could matter, and the answers dif-  
943 fer. The first is learning the crop’s response to shade from data. The learned  
944 model is useful mainly as a measurement of how little can be learned: a hier-  
945 archical model fitted to 428 observations from 58 studies predicts a held-out  
946 study of the four staple crop types with an error of 0.22 in relative yield, no  
947 better than proportional loss, and the gain over proportional loss (0.43 to  
948 0.31 over all types) comes from crop types that the simulator does not repre-  
949 sent. Crop type is the information that transfers, whereas irradiation, climate  
950 zone and shade type add nothing measurable and a non-parametric model  
951 (gradient-boosted trees) is worse with this number of studies (Section 3.7).  
952 Two things follow. The shade response of the staple crops is a distribution  
953 with a spread of about 20 percentage points between studies and not a curve,  
954 and the process-based simulator does not reproduce the steep losses of maize  
955 and legumes seen in the data. Learning the response was limited by the data  
956 and not by the model: two studies support the tuber curve, no rice study  
957 exists, and 46 panel observations are all leafy vegetables.

958 The second is calibration through the differentiable simulator. Gradient  
959 descent through whole seasons fitted the response of each crop to the learned  
960 curves and to the field trials, and it produced an ensemble of plausible re-  
961 sponses that reconciles them (Section 3.8). The ensemble informs the design  
962 decision: a density chosen under the mean response leaves about 70% of  
963 crop–site pairs at the floor if another plausible response holds, against 85–  
964 87% for an oracle, and a density chosen to meet the floor in 75% of plausible  
965 responses recovers about half of that gap at a cost of 0.04 LER. A simple mar-  
966 gin, however, does as well: raising the floor from 0.90 to 0.92 under the mean  
967 response reaches 85–88% of the pairs, and interpolating the margin frontier  
968 at the compliance of the ensemble rules gives LER differences (margin minus  
969 rule) of +0.001 (−0.009 to 0.012) and +0.003 (−0.016 to 0.021) under E2  
970 and −0.006 (−0.019 to 0.005) and +0.024 (−0.011 to 0.058) under E3, so the  
971 two are not distinguishable on this sample, and the cost per point of com-  
972 pliance is similar. None of the rules is safe outside the ensemble (7–33% of  
973 the pairs at the floor under the proportional, calibrated and harsh responses  
974 for the mean design, the margin of 0.92 and the ensemble rules), and only a  
975 floor of 0.96 reaches 51–78% at a cost of about 0.2 LER. What the ensem-  
976 ble adds is the size of the margin, not a better rule. The quantile rule is a  
977 scenario-type design (Calafiore and Campi, 2006; Campi and Garatti, 2008);  
978 with five members it has none of the finite-sample guarantees of that theory,

979 and the comparison with the margin shows that it is not needed in practice.  
980 The bottleneck of tracker-based agrivoltaics under a yield-retention floor is  
981 the uncertainty of the crop response that enters the design, and it is data  
982 on that response, not larger networks, that would improve the result; the  
983 learned curves show a spread of about 20 percentage points between studies.  
984 The guarantee holds only within the range of responses that the ensemble  
985 spans.

986 The third is control, and here learning contributes little. Differentia-  
987 bility is what lets a network of 5441 weights be trained in 300 gradient  
988 steps through whole seasons; a derivative-free search such as CMA-ES han-  
989 dled a five-parameter schedule with a few hundred simulator evaluations and  
990 reached the performance of the tuned window, but would not scale to the net-  
991 work, and the network’s remaining margin over it is small (0.013 LER). The  
992 learned controller adds 0.018 LER over a tuned phenology rule (and 0.044  
993 over the written one), and 0.015–0.038 over the written rule under the four  
994 alternative responses on one split (21 pairs); a reader who needs a simple,  
995 auditable controller can use that rule and obtain most of the electricity gain  
996 over static arrays. Neither of the two extensions that we expected to matter  
997 under uncertainty did: training on ensembles of responses (E1, E2 and E3)  
998 did not help and, for the wider ensembles, did not retain the margin over the  
999 written rule within our training budget (one seed, 21 pairs), and neither did  
1000 giving the controller a farm-measurable signal, the biomass of the crop rela-  
1001 tive to a reference plot, nor a feedback rule built on that signal. The learned  
1002 policy also has to be initialised from the phenology rule (Section 4.5). We  
1003 therefore do not claim that learned control is what makes agrivoltaics work:  
1004 in this model, the schedule that a learned policy converges to is a smooth  
1005 function of crop phenology, which a rule can already express.

#### 1006 4.3. *What the results do not establish*

1007 The early hypothesis that stress-aware control, that is, shading the crop  
1008 on heat and drought days, would be a main source of benefit under warming  
1009 was not supported. The stress rule performs worse than the phenology rule  
1010 in every climate, and a learned controller without stress inputs performs as  
1011 well as one with them. A plausible reason is that in our crop model the yield  
1012 gain from shading on a hot or dry day is smaller than the loss of light on  
1013 that day, so stress information does not help. Whether this holds for crop  
1014 models with more detailed canopy-temperature and heat-stress responses is  
1015 an open question. Agrivoltaics does not protect staple production against

warming in our simulations: light-sharing systems produce about 90% of the open-field yield in every climate and follow the same decline in maize and rice (Fig. 8).

#### 4.4. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.11) shows the trade-off in magnitude. The 14 PWh yr<sup>-1</sup> that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

**Who bears the risk.** A floor defined on the multi-year mean leaves the farmer exposed in the seasons below it: with the density designed for the floor, about 30% of seasons fall short on the main split (70% compliance for SALS), and a per-season guarantee through chance-constrained design costs 0.065 LER. Siting by electricity favours maize and soybean areas, where compliance is lowest (Section 3.11), so the risk concentrates where the energy gain is largest. Who carries it, through contracts, insurance or a premium, is a design question that the model does not address, and food is valued only through crop revenue at market prices.

#### 4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the trials (rice, wheat, potato, maize) but too mild for maize and soybean relative to the meta-analysis, where the 40%-shading relative yields are 0.42 and 0.40 against 0.75 and 0.48 in the simulator. The harvest-index sensitivities  $h$  are assumptions, and the sensitivity analyses use subsamples of crop-site

1051 pairs (22 for the rule comparison under alternative responses, 21 for  
1052 the controller comparison, one training seed).

- 1053 • *Canopy cooling.* The effect of shade on canopy temperature was repre-  
1054 sented by one coefficient  $\kappa$ , applied to a heat-stress function that was  
1055 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).  
1056 The ranking of the controllers is robust to  $\kappa$  between 0 and 2 °C, but  
1057 the absolute electricity changes by about  $\pm 8\%$ . Nocturnal warming  
1058 under modules and rain redistribution are not modelled.
- 1059 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-  
1060 use efficiency, and its CO<sub>2</sub> response is that of controlled experiments  
1061 that tend to overstate field responses (Long et al., 2006). More detailed  
1062 models with grain-number formation and canopy energy balance may  
1063 favour different schedules.
- 1064 • *Layouts and models.* The baseline of static arrays is a backtracking  
1065 tracker held at the energy-optimal angle; flat and 30°-tilted fixed ar-  
1066 rays did no better in a check on 21 pairs, but vertical bifacial layouts  
1067 (Willockx et al., 2023; Campana et al., 2021) are not modelled, and the  
1068 ground coverage ratio is a coarse proxy of crop productivity (Dupraz,  
1069 2023). The PV model is isotropic in the diffuse component and has no  
1070 bifacial gain; it was benchmarked against PVGIS at low density only  
1071 and has not been compared with a reference implementation such as  
1072 pvlib (F. Holmgren et al., 2018). The rotation limit was varied between  
1073 45 and 80 degrees and changes the size of the benefit but not the rank-  
1074 ing; the harvest-index window is assumed to be common to all crops,  
1075 and shifting it or removing the penalty did not change the ranking of  
1076 the rules.
- 1077 • *Climate scenarios.* Warming was applied as uniform temperature shifts  
1078 with fixed precipitation, radiation and relative humidity, and with un-  
1079 changed cultivars and sowing dates. It is not a bias-adjusted climate  
1080 projection.
- 1081 • *Initialisation and state information.* The network is initialised from the  
1082 phenology rule, because training from random weights collapsed to full  
1083 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,  
1084 and we cannot say whether a better optimiser or a richer policy class

1085 would find larger gains. The controller uses the simulated state of the  
1086 crop and soil (thermal time, canopy interception, soil water) without  
1087 measurement error, and the forecast errors we tested apply to weather  
1088 only.

- 1089 • *Economics and deployment.* The economic analysis (Section 3.9) uses  
1090 assumed prices and costs, no grid, curtailment, storage or subsidy  
1091 model, and does not value food security; its break-even prices move  
1092 with the capex and the discount rate (Fig. 7). The value of midday  
1093 electricity, curtailment, storage, structural cost and farmer acceptance  
1094 (Trommsdorff et al., 2021) will determine the realisable share. The av-  
1095 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)  
1096 leaves room for machinery, but the densest designs (GCR up to 0.5)  
1097 have a pitch of only 4.4 m.

## 1098 5. Conclusions

1099 Light sharing, the daily choice between maximum electricity and letting  
1100 light through to the crop, is the main lever on the electricity of tracker-based  
1101 agrivoltaics at a given yield-retention floor, although it pays only at high  
1102 electricity prices: below about 80 USD MWh<sup>-1</sup> static arrays earn more. In  
1103 a differentiable simulator of trackers, PV output, crop light, microclimate  
1104 and crop growth, evaluated on held-out crop-site pairs (92 on the main split,  
1105 76 distinct pairs pooled over three splits for the controller and design com-  
1106 parisons) with the array density of each site chosen from its own history,  
1107 light-sharing rules allowed 2.1 to 2.9 times the module density of static ar-  
1108 rays at a mean yield retention of 90%. They raised the land-equivalent ratio  
1109 from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no  
1110 weather forecast was the best of them. The simulated shade response is con-  
1111 servative against the field trials we could use for rice, wheat and potato, and  
1112 calibrating it raised the gains without changing the ranking of methods; for  
1113 maize and soybean, which supply 53% of the upscaled electricity, the meta-  
1114 analysis indicates larger losses than the simulator reproduces whereas one  
1115 maize trial indicates smaller ones, so the gains for these crops are uncertain  
1116 and possibly overestimated. The crop’s response to shade, the least certain  
1117 input, was learned from 428 field observations: a hierarchical model pre-  
1118 dicted a held-out study of the staple crop types no better than proportional  
1119 loss (error 0.22), which shows the size of the between-study spread, and gra-  
1120 dient descent through the simulator turned it into an ensemble of plausible

1121 responses. A learned controller exceeded the written phenology rule by 0.044  
 1122 LER and a rule with a window tuned on the training seasons by 0.018 (95%  
 1123 interval 0.012–0.024), by following a smooth phenology-dependent schedule  
 1124 that a rule can express; training it on ensembles of shade responses or giv-  
 1125 ing it a farm-measurable biomass signal did not help, and with the wider  
 1126 ensembles it did not retain its margin over the written rule in our training  
 1127 budget (one seed, 21 pairs). The decision that mattered was the design. A  
 1128 density chosen under the mean response left about 70% of crop–site pairs  
 1129 at the floor when another plausible response held (oracle 85–87%); choosing  
 1130 the density so that the floor holds in 75% of the plausible responses recov-  
 1131 ered about half of the gap at a cost of 0.04 LER, but a floor raised by 0.02  
 1132 under the mean response performed alike at equal compliance (differences  
 1133 of  $-0.006$  to  $+0.024$  LER with intervals containing zero), and no rule was  
 1134 safe outside the ensemble. The ensemble is therefore useful to size a safety  
 1135 margin, not to replace it. Under illustrative costs, light sharing breaks even  
 1136 with the open field at 70 to 92 USD MWh<sup>-1</sup> (73 for the learned controller,  
 1137 59 for static arrays); a conventional plant replacing the crop is more prof-  
 1138 itable above its break-even of 48 USD MWh<sup>-1</sup>, so agrivoltaics depends on a  
 1139 premium, land-use rules or a value for food production that markets do not  
 1140 give. The yield floor is met on average but not in every season or region,  
 1141 and the electricity potential of 5% of staple cropland (about 14 PWh yr<sup>-1</sup>)  
 1142 requires about 10 TWp of modules. Future work should validate the crop  
 1143 response to shade under trackers with more field data, notably for maize and  
 1144 legumes where the meta-analysis and the trials disagree, extend the crop  
 1145 model to grain-number formation and canopy energy balance, add costs and  
 1146 grid constraints, and pursue controllers that guarantee the yield-retention  
 1147 floor per site and per season.

#### 1148 **CRedit authorship contribution statement**

1149 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal  
 1150 analysis, Investigation, Data curation, Visualization, Writing – original draft,  
 1151 Writing – review & editing.

#### 1152 **Declaration of competing interest**

1153 The author has no known competing financial interests or personal re-  
 1154 lationships that could have appeared to influence the work reported in this  
 1155 paper.

## **Data and code availability**

All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al., 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author’s public repository (<https://github.com/vinhq dang/Artificial-Intelligence-in-Agriculture>).

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## **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling  $\kappa = 1^\circ\text{C}$ ). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is  $\geq 0.9$ .  $E$ : electricity ( $\text{MWh ha}^{-1} \text{ yr}^{-1}$ );  $e$ : electricity relative to a conventional PV plant;  $r$ : mean yield retention;  $r_{\text{prod}}$ : production-weighted retention;  $C_s$  and  $C_{\text{site}}$ : share of seasons with  $r \geq 0.9$  and share of sites whose test-season mean  $r \geq 0.9$  (%);  $\text{LER} = r + e$ .

| Method           | GCR  | $E$ | $e$  | $r$   | $r_{\text{prod}}$ | $C_s$ | $C_{\text{site}}$ | LER  |
|------------------|------|-----|------|-------|-------------------|-------|-------------------|------|
| <i>Baseline</i>  |      |     |      |       |                   |       |                   |      |
| AV static        | 0.06 | 217 | 0.16 | 0.917 | 0.914             | 93    | 96                | 1.07 |
| Seasonal sharing | 0.17 | 397 | 0.28 | 0.904 | 0.902             | 69    | 72                | 1.19 |
| Phenology rule   | 0.17 | 465 | 0.33 | 0.904 | 0.902             | 69    | 75                | 1.24 |
| Stress rule      | 0.13 | 387 | 0.28 | 0.907 | 0.907             | 76    | 82                | 1.18 |
| SALS (ours)      | 0.17 | 510 | 0.36 | 0.905 | 0.902             | 70    | 74                | 1.27 |
| <i>+1.5 °C</i>   |      |     |      |       |                   |       |                   |      |
| AV static        | 0.06 | 221 | 0.16 | 0.916 | 0.914             | 93    | 92                | 1.08 |
| Seasonal sharing | 0.16 | 393 | 0.28 | 0.905 | 0.903             | 72    | 74                | 1.19 |
| Phenology rule   | 0.17 | 478 | 0.34 | 0.905 | 0.902             | 68    | 75                | 1.25 |
| Stress rule      | 0.12 | 377 | 0.27 | 0.908 | 0.908             | 73    | 77                | 1.18 |
| SALS (ours)      | 0.17 | 518 | 0.37 | 0.906 | 0.903             | 68    | 75                | 1.28 |
| <i>+2 °C</i>     |      |     |      |       |                   |       |                   |      |
| AV static        | 0.06 | 223 | 0.16 | 0.916 | 0.914             | 92    | 93                | 1.08 |
| Seasonal sharing | 0.17 | 394 | 0.28 | 0.905 | 0.904             | 72    | 76                | 1.19 |
| Phenology rule   | 0.17 | 483 | 0.35 | 0.905 | 0.903             | 69    | 74                | 1.25 |
| Stress rule      | 0.12 | 370 | 0.26 | 0.909 | 0.909             | 75    | 78                | 1.17 |
| SALS (ours)      | 0.17 | 521 | 0.38 | 0.907 | 0.904             | 69    | 75                | 1.28 |
| <i>+3 °C</i>     |      |     |      |       |                   |       |                   |      |
| AV static        | 0.06 | 231 | 0.17 | 0.915 | 0.914             | 88    | 89                | 1.08 |
| Seasonal sharing | 0.17 | 397 | 0.29 | 0.906 | 0.904             | 69    | 75                | 1.19 |
| Phenology rule   | 0.18 | 499 | 0.36 | 0.905 | 0.902             | 67    | 72                | 1.26 |
| Stress rule      | 0.11 | 366 | 0.26 | 0.908 | 0.908             | 74    | 77                | 1.17 |
| SALS (ours)      | 0.18 | 540 | 0.39 | 0.907 | 0.904             | 68    | 75                | 1.30 |

Table 3: LER of the controllers at different yield-retention floors  $\rho$  (main split; the historical design step is repeated for each floor).

| Method                          | Baseline climate |       |       |       | +3 °C |       |       |       |
|---------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                                 | 0.80             | 0.85  | 0.90  | 0.95  | 0.80  | 0.85  | 0.90  | 0.95  |
| AV static                       | 1.159            | 1.115 | 1.072 | 1.011 | 1.174 | 1.128 | 1.083 | 1.013 |
| Stress rule                     | 1.380            | 1.282 | 1.182 | 1.077 | 1.364 | 1.269 | 1.172 | 1.071 |
| Phenology rule                  | 1.467            | 1.356 | 1.236 | 1.108 | 1.513 | 1.395 | 1.265 | 1.122 |
| SALS (trained at $\rho = 0.9$ ) | 1.505            | 1.392 | 1.270 | 1.141 | 1.553 | 1.432 | 1.297 | 1.154 |
| SALS conditioned on $\rho$      | 1.504            | 1.391 | 1.269 | 1.141 | 1.553 | 1.431 | 1.296 | 1.154 |

Table 4: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

| Setting                            | Static tracker | Phenology rule | Tuned rule | SALS  |
|------------------------------------|----------------|----------------|------------|-------|
| Main (limit 60°, window 0.45–0.65) | 1.053          | 1.170          | 1.180      | 1.189 |
| Rotation limit 45°                 | 1.057          | 1.138          | 1.144      | 1.144 |
| Rotation limit 80°                 | 1.053          | 1.206          | 1.219      | 1.224 |
| Window 0.30–0.50                   | 1.053          | 1.169          | 1.176      | 1.185 |
| Window 0.60–0.80                   | 1.053          | 1.171          | 1.181      | 1.182 |
| No harvest-index penalty           | 1.109          | 1.328          | 1.347      | 1.353 |

Table 5: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons of split 0; the written rule uses 0.05 to 1.0). The optimised ramp schedule has five parameters (start 0.17, end 0.93, ramp widths 0.02, level 0.99) found by CMA-ES on the same training seasons.

| Method                  | GCR   | $E/E_{PV}$ | Retention | Seasons $\geq 0.9$ (%) | Pairs $\geq 0.9$ (%) | LER   |
|-------------------------|-------|------------|-----------|------------------------|----------------------|-------|
| AV static               | 0.059 | 0.153      | 0.922     | 95                     | 96                   | 1.076 |
| Seasonal sharing        | 0.151 | 0.249      | 0.915     | 84                     | 91                   | 1.164 |
| Stress rule             | 0.118 | 0.255      | 0.918     | 83                     | 88                   | 1.173 |
| Phenology rule          | 0.154 | 0.292      | 0.916     | 85                     | 91                   | 1.207 |
| Tuned phenology rule    | 0.161 | 0.317      | 0.916     | 84                     | 89                   | 1.233 |
| Optimised ramp schedule | 0.155 | 0.322      | 0.916     | 83                     | 87                   | 1.238 |
| SALS                    | 0.161 | 0.335      | 0.916     | 83                     | 87                   | 1.251 |

Table 6: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

| Trial                                     | Shading | Observed | Conservative | Calibrated |
|-------------------------------------------|---------|----------|--------------|------------|
| Rice, Gonocruz et al. (2021)              | 0.25    | 0.84     | 0.77         | 0.79       |
| Wheat, Weselek et al. (2021)              | 0.30    | 0.81     | 0.66         | 0.74       |
| Potato, Weselek et al. (2021)             | 0.30    | 0.82     | 0.72         | 0.72       |
| Maize, Ramos-Fuentes et al. (2023) (DAV)  | 0.35    | 0.83     | 0.70         | 0.70       |
| Maize, Ramos-Fuentes et al. (2023) (half) | 0.30    | 0.91     | 0.75         | 0.75       |
| Maize, Ramos-Fuentes et al. (2023) (full) | 0.50    | 0.70     | 0.54         | 0.54       |

Table 7: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor  $\rho = 0.9$ , split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

| Shade response | Method           | GCR  | $E/E_{PV}$ | Retention | LER  |
|----------------|------------------|------|------------|-----------|------|
| conservative   | AV static        | 0.06 | 0.15       | 0.92      | 1.07 |
|                | Phenology rule   | 0.17 | 0.33       | 0.90      | 1.23 |
|                | Seasonal sharing | 0.16 | 0.28       | 0.90      | 1.18 |
|                | Stress rule      | 0.11 | 0.24       | 0.91      | 1.15 |
| calibrated     | AV static        | 0.07 | 0.18       | 0.91      | 1.09 |
|                | Phenology rule   | 0.19 | 0.37       | 0.90      | 1.28 |
|                | Seasonal sharing | 0.19 | 0.32       | 0.90      | 1.22 |
|                | Stress rule      | 0.13 | 0.29       | 0.91      | 1.20 |
| field-fitted   | AV static        | 0.16 | 0.41       | 0.91      | 1.32 |
|                | Phenology rule   | 0.37 | 0.70       | 0.92      | 1.62 |
|                | Seasonal sharing | 0.37 | 0.61       | 0.92      | 1.53 |
|                | Stress rule      | 0.29 | 0.63       | 0.90      | 1.53 |

Table 8: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with  $s > 0$ ). Covariates: irradiation, climate zone, shade type, intercropping.

| Model                                | RMSE  | MAE   | Median AE | Mean MAE per study |
|--------------------------------------|-------|-------|-----------|--------------------|
| Proportional loss, $y = 1 - s$       | 0.430 | 0.305 | 0.210     | 0.224              |
| One curve for all crops              | 0.372 | 0.265 | 0.193     | 0.208              |
| Curve per crop type, no pooling      | 0.313 | 0.208 | 0.141     | 0.165              |
| Curve per crop type, partial pooling | 0.312 | 0.210 | 0.150     | 0.162              |
| Partial pooling + covariates         | 0.311 | 0.206 | 0.148     | 0.161              |
| Gradient-boosted trees               | 0.321 | 0.219 | 0.162     | 0.184              |

Table 9: Leave-one-study-out prediction error (RMSE of relative yield, observations with  $s > 0$ ) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

| Group                                              | Obs./studies | Proportional | Hierarchical | Boosted t |
|----------------------------------------------------|--------------|--------------|--------------|-----------|
| C3 cereals                                         | 30/10        | 0.214        | 0.187        | 0.205     |
| Maize                                              | 40/10        | 0.225        | 0.235        | 0.248     |
| Grain legumes                                      | 58/14        | 0.215        | 0.225        | 0.234     |
| Tubers/root crops                                  | 10/2         | 0.250        | 0.254        | 0.257     |
| Four crop types used by the simulator              | 138/30       | 0.220        | 0.222        | 0.234     |
| Other types (berries, forages, fruits, vegetables) | 201/29       | 0.528        | 0.361        | 0.369     |
| All types                                          | 339/58       | 0.430        | 0.312        | 0.321     |

Table 10: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

| Crop    | Shading | Data-driven       | Simulator, E1 | Simulator, E2 | Proportional |
|---------|---------|-------------------|---------------|---------------|--------------|
| Wheat   | 20%     | 0.93 [0.88, 0.96] | 0.88          | 0.92          | 0.80         |
| Wheat   | 40%     | 0.72 [0.59, 0.78] | 0.70          | 0.77          | 0.60         |
| Rice    | 20%     | 0.93 [0.88, 0.96] | 0.85          | 0.90          | 0.80         |
| Rice    | 40%     | 0.72 [0.59, 0.78] | 0.66          | 0.73          | 0.60         |
| Maize   | 20%     | 0.73 [0.58, 0.86] | 0.86          | 0.90          | 0.80         |
| Maize   | 40%     | 0.42 [0.27, 0.59] | 0.69          | 0.75          | 0.60         |
| Soybean | 20%     | 0.71 [0.65, 0.79] | 0.77          | 0.74          | 0.80         |
| Soybean | 40%     | 0.40 [0.34, 0.53] | 0.53          | 0.48          | 0.60         |
| Potato  | 20%     | 0.93 [0.92, 0.97] | 0.88          | 0.92          | 0.80         |
| Potato  | 40%     | 0.71 [0.67, 0.83] | 0.73          | 0.79          | 0.60         |

Table 11: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean  $\pm$  s.d.): radiation-use compensation  $c$  and multiplier  $h$  on the flowering-window harvest-index penalty (1 = assumed value).

| Crop    | $c$             | $h$ multiplier  |
|---------|-----------------|-----------------|
| Wheat   | $0.91 \pm 0.31$ | $0.74 \pm 0.53$ |
| Rice    | $0.61 \pm 0.25$ | $0.83 \pm 0.33$ |
| Maize   | $0.14 \pm 0.11$ | $0.16 \pm 0.13$ |
| Soybean | $0.05 \pm 0.09$ | $1.66 \pm 0.38$ |
| Potato  | $0.62 \pm 0.20$ | $0.90 \pm 0.50$ |

Table 12: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor  $\rho = 0.9$ , every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

| Method                  | Proportional | Calibrated | Field-fitted | Harsh |
|-------------------------|--------------|------------|--------------|-------|
| AV static               | 1.052        | 1.059      | 1.254        | 1.037 |
| Seasonal sharing        | 1.133        | 1.156      | 1.385        | 1.097 |
| Stress rule             | 1.129        | 1.160      | 1.406        | 1.086 |
| Feedback rule           | 1.139        | 1.164      | 1.392        | 1.089 |
| Phenology rule          | 1.164        | 1.190      | 1.456        | 1.122 |
| Tuned phenology rule    | 1.170        | 1.199      | 1.472        | 1.129 |
| Optimised ramp schedule | 1.170        | 1.199      | 1.477        | 1.122 |
| SALS                    | 1.178        | 1.209      | 1.489        | 1.132 |
| SALS-robust, E1         | 1.179        | 1.204      | 1.491        | 1.127 |
| SALS-adaptive, E1       | 1.176        | 1.202      | 1.495        | 1.124 |
| SALS-robust, E2         | 1.165        | 1.192      | 1.483        | 1.126 |
| SALS-robust, E3         | 1.163        | 1.191      | 1.481        | 1.121 |
| SALS-adaptive, E3       | 1.164        | 1.191      | 1.475        | 1.118 |

Table 13: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop-site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention  $\geq 0.9$ ; shortfall: mean of  $\max(0, 0.9 - \text{retention})$  per pair. The LER values exceed those of the main analysis because E2 and E3 are milder than the proportional response and allow denser arrays; the margin grid (0.92, 0.94, 0.96) was chosen after a first analysis of E2 and is not pre-specified.

| Design                            | E2                   |           |       | E3                   |           |       |
|-----------------------------------|----------------------|-----------|-------|----------------------|-----------|-------|
|                                   | Pairs $\geq 0.9$ (%) | Shortfall | LER   | Pairs $\geq 0.9$ (%) | Shortfall | LER   |
| oracle (true response)            | 87                   | 0.0025    | 1.486 | 85                   | 0.0029    | 1.486 |
| mean response                     | 70                   | 0.0062    | 1.493 | 71                   | 0.0093    | 1.493 |
| floor met in 50% of other members | 65                   | 0.0084    | 1.510 | 67                   | 0.0125    | 1.510 |
| floor met in 75% of other members | 78                   | 0.0045    | 1.462 | 79                   | 0.0083    | 1.462 |
| floor met in all other members    | 89                   | 0.0019    | 1.412 | 88                   | 0.0032    | 1.412 |
| floor 0.92 under mean response    | 88                   | 0.0022    | 1.426 | 85                   | 0.0051    | 1.426 |
| floor 0.94 under mean response    | 97                   | 0.0007    | 1.345 | 92                   | 0.0027    | 1.345 |
| floor 0.96 under mean response    | 99                   | 0.0003    | 1.251 | 96                   | 0.0015    | 1.251 |

Table 14: Design rules outside the ensemble: share of crop-site pairs (%) with mean retention  $\geq 0.9$  under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). Designs use the whole ensemble E2 or E3 (five members and their mean); the oracle uses the true response. LER in the last column is the mean over the four responses.

| Design                         | E2 |    |    |    | E3 |    |    |    | LER   |       |
|--------------------------------|----|----|----|----|----|----|----|----|-------|-------|
|                                | P  | C  | F  | H  | P  | C  | F  | H  | E2    | E3    |
| oracle (true response)         | 91 | 89 | 87 | 84 | 91 | 89 | 87 | 84 | 1.283 | 1.283 |
| mean response                  | 13 | 13 | 68 | 12 | 13 | 17 | 79 | 7  | 1.435 | 1.448 |
| floor met in 75% of members    | 16 | 16 | 80 | 12 | 22 | 25 | 93 | 13 | 1.384 | 1.397 |
| floor met in all members       | 17 | 17 | 80 | 12 | 29 | 32 | 93 | 16 | 1.348 | 1.336 |
| floor 0.92 under mean response | 17 | 20 | 86 | 12 | 29 | 33 | 91 | 20 | 1.376 | 1.399 |
| floor 0.94 under mean response | 45 | 47 | 97 | 34 | 45 | 45 | 97 | 29 | 1.303 | 1.345 |
| floor 0.96 under mean response | 63 | 78 | 99 | 54 | 58 | 63 | 99 | 51 | 1.220 | 1.264 |

Table 15: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

| Parameter                           | Value                                                                    |
|-------------------------------------|--------------------------------------------------------------------------|
| Electricity price                   | 60 USD MWh <sup>-1</sup> (30–120)                                        |
| Capex, AV single-axis tracker plant | 1.0 USD Wp <sup>-1</sup> (0.7–1.5)                                       |
| Capex, conventional tracker plant   | 0.75 USD Wp <sup>-1</sup>                                                |
| Discount rate, lifetime, O&M        | 6% (4–8%), 25 years, 1.5% of capex per year                              |
| Installed capacity                  | 2.1 MWp per hectare of module area share $g$ (0.21 kWp m <sup>-2</sup> ) |
| Crop price per tonne of dry matter  | wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.         |

Table 16: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 15). Net value is relative to the open field (crop costs unchanged) in kUSD ha<sup>-1</sup> yr<sup>-1</sup> at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

| Method                | $E$ (MWh) | Capex annuity | Crop loss | Net at 60 | Net at 90 | Net at 120 |
|-----------------------|-----------|---------------|-----------|-----------|-----------|------------|
| AV static             | 200       | 11.5          | 0.27      | 0.3       | 6.3       | 12.3       |
| Seasonal sharing      | 326       | 29.6          | 0.31      | -10.3     | -0.5      | 9.2        |
| Stress rule           | 336       | 23.2          | 0.30      | -3.3      | 6.8       | 16.9       |
| Phenology rule        | 382       | 30.1          | 0.30      | -7.5      | 4.0       | 15.4       |
| Tuned phenology rule  | 415       | 31.6          | 0.31      | -7.0      | 5.5       | 17.9       |
| SALS                  | 438       | 31.6          | 0.31      | -5.6      | 7.6       | 20.7       |
| Conventional PV plant | 1306      | 58.7          | 3.60      | 16.0      | 55.2      | 94.4       |

Table 17: One-at-a-time sensitivity of the net value (kUSD ha<sup>-1</sup> yr<sup>-1</sup>, relative to the open field) at 90 USD MWh<sup>-1</sup>, densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

| Parameter         | Value | Static arrays | Tuned rule | SALS | Conventional plant |
|-------------------|-------|---------------|------------|------|--------------------|
| Base              |       | 6.3           | 5.5        | 7.6  | 55.2               |
| AV capex (USD/Wp) | 0.7   | 9.7           | 14.9       | 17.1 | 72.8               |
| AV capex (USD/Wp) | 1.5   | 0.6           | -10.3      | -8.2 | 25.8               |
| Discount rate     | 4%    | 8.0           | 10.3       | 12.4 | 64.1               |
| Discount rate     | 8%    | 4.4           | 0.2        | 2.4  | 45.4               |
| Crop prices       | 0.5×  | 6.4           | 5.6        | 7.7  | 57.0               |
| Crop prices       | 2.0×  | 6.0           | 5.2        | 7.3  | 51.6               |
| Electricity price | 60    | 0.3           | -7.0       | -5.6 | 16.0               |
| Electricity price | 120   | 12.3          | 17.9       | 20.7 | 94.4               |

Table A.18: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons).  $\Delta\text{LER}$ : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI;  $p$ : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER;  $\Delta r$ : difference in mean yield retention with its Wilcoxon  $p$ -value.

| Climate  | Baseline         | $\Delta\text{LER}$ [95% CI] | $p$         | % better | $\Delta r$ | $p$         | Sites |
|----------|------------------|-----------------------------|-------------|----------|------------|-------------|-------|
| Baseline | AV static        | +0.205 [+0.176, +0.236]     | $< 10^{-4}$ | 100      | -0.011     | $< 10^{-4}$ | 92    |
| Baseline | Seasonal sharing | +0.088 [+0.068, +0.109]     | $< 10^{-4}$ | 97       | +0.001     | 0.922       | 92    |
| Baseline | Phenology rule   | +0.038 [+0.025, +0.053]     | $< 10^{-4}$ | 84       | +0.001     | 0.723       | 92    |
| Baseline | Stress rule      | +0.096 [+0.071, +0.124]     | $< 10^{-4}$ | 93       | -0.002     | 0.055       | 92    |
| +1.5 °C  | AV static        | +0.208 [+0.180, +0.237]     | $< 10^{-4}$ | 100      | -0.010     | $< 10^{-4}$ | 92    |
| +1.5 °C  | Seasonal sharing | +0.095 [+0.077, +0.114]     | $< 10^{-4}$ | 98       | +0.001     | 0.812       | 92    |
| +1.5 °C  | Phenology rule   | +0.033 [+0.022, +0.045]     | $< 10^{-4}$ | 86       | +0.002     | 0.855       | 92    |
| +1.5 °C  | Stress rule      | +0.108 [+0.083, +0.134]     | $< 10^{-4}$ | 98       | -0.001     | 0.041       | 92    |
| +2 °C    | AV static        | +0.209 [+0.181, +0.238]     | $< 10^{-4}$ | 100      | -0.010     | $< 10^{-4}$ | 92    |
| +2 °C    | Seasonal sharing | +0.097 [+0.079, +0.115]     | $< 10^{-4}$ | 97       | +0.001     | 0.876       | 92    |
| +2 °C    | Phenology rule   | +0.032 [+0.022, +0.044]     | $< 10^{-4}$ | 85       | +0.002     | 0.496       | 92    |
| +2 °C    | Stress rule      | +0.115 [+0.089, +0.142]     | $< 10^{-4}$ | 98       | -0.003     | 0.016       | 92    |
| +3 °C    | AV static        | +0.219 [+0.190, +0.250]     | $< 10^{-4}$ | 100      | -0.009     | $< 10^{-4}$ | 92    |
| +3 °C    | Seasonal sharing | +0.108 [+0.089, +0.129]     | $< 10^{-4}$ | 96       | +0.001     | 0.935       | 92    |
| +3 °C    | Phenology rule   | +0.034 [+0.022, +0.046]     | $< 10^{-4}$ | 86       | +0.002     | 0.330       | 92    |
| +3 °C    | Stress rule      | +0.131 [+0.102, +0.162]     | $< 10^{-4}$ | 98       | -0.002     | 0.142       | 92    |